LOCAL-PROMPT: EXTENSIBLE LOCAL PROMPTS FOR FEW-SHOT OUT-OF-DISTRIBUTION DETECTION

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ABSTRACT

Few-shot out-of-distribution (OOD) detection with vision–language models (VLMs) is challenging because hard OOD samples often differ from in-distribution (ID) classes only in subtle, localized regions. We propose Local-Prompt, a coarse-to-fine prompting framework that freezes global prompts and learns class-specific local prompts to capture regional outlier cues. The method has two stages: (1) global-prompt-guided negative augmentation that uses random crops and frozen global prompts to select hard positives and negatives without requiring real OOD data; and (2) local-prompt training with regional regularization using top-k regions, a bank of local negative prompts, and a diversity loss on negatives. At inference, a Regional-MCM score aggregates global Maximum Concept Matching (MCM) with top-k regional scores, including negative prompts, yielding a local-aware OOD decision and an ID score. On ImageNet-1k (4-shot), Local-Prompt combined with LoCoOp reduces average FPR95 from 31.46 to 26.29 and increases AUROC from 92.68 to 94.03, surpassing prior 16-shot methods. Local-Prompt alone (4-shot) improves over CoOp (FPR95 27.66 vs 33.82; AUROC 93.53 vs 90.98). It also improves ID accuracy by +1.02% (4-shot) and +2.38% (16-shot) over LoCoOp. On ImageNet-100 (4-shot), Local-Prompt achieves average FPR95 23.42 and AUROC 95.81, outperforming MCM (32.58/94.48) and LoCoOp (41.47/92.99). For near"-OOD (ImageNet-10/20, 16-shot), it attains FPR95 5.05 and AUROC 98.95, markedly better than GL-MCM (9.55/98.33) and LoCoOp (11.60/97.64). These results demonstrate that focusing prompt learning on regional cues with hard negatives and a regional OOD score advances few-shot OOD detection while remaining extensible to stronger global prompt designs.

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1 INTRODUCTION

Out-of-distribution (OOD) detection aims to identify inputs that do not belong to the training distribution, a core requirement for deploying reliable machine learning systems. Recent vision–language models (VLMs) such as CLIP (Radford et al., 2021) enable zero-shot recognition by aligning images and texts in a shared space. However, in few-shot settings where only a handful of labeled ID samples per class are available, OOD detection remains difficult, especially for "near"""" cases where OOD images differ from ID classes only in subtle, localized regions. Global prompt learning methods like CoOp (Zhou et al., 2022) and LoCoOp (Miyai et al., 2023) improve ID classification but can overlook fine-grained regional cues crucial for separating hard OOD samples.

We hypothesize that local regions carry discriminative signals for separating near" and subtle OOD cases that global features miss. We operationalize this through Local-Prompt, which decomposes prompting into global (frozen) and local (trainable) prompts, and adopts a two-stage scheme: (1) global prompt guided negative augmentation, which applies random crops and uses frozen global prompts to select the hardest negatives/positives, injecting local outlier knowledge without real OOD data; and (2) local prompt enhanced regional regularization, which trains class-specific local prompts and a bank of local negative prompts via contrastive learning over top-k regions together with a diversity loss on negatives. A Regional-MCM score aggregates global MCM (Ming et al., 2022) with top-k regional scores (including negative prompts), producing a local-aware OOD score and an ID score.

Empirically, Local-Prompt consistently improves few-shot OOD detection across ImageNet-1k and ImageNet-100 benchmarks, and further boosts ID accuracy. Notably, on ImageNet-1k with 4 shots per class, combining Local-Prompt with LoCoOp reduces average FPR95 by 5.17 points and improves AUROC by 1.35 (Table 1); on ImageNet-100 (4-shot) it substantially improves over strong baselines (Table 4). The approach also excels on near"-OOD detection (ImageNet-10/20; **??**).

Our contributions are threefold: (1) a coarse-to-fine few-shot paradigm that freezes global prompts and trains local prompts to capture regional outlier cues; (2) global prompt guided hard negative augmentation and regional regularization (contrastive top-k regions and diversity) with local negative prompts; and (3) a Regional-MCM score that combines global and local evidence for robust OOD detection and classification, with seamless extensibility to plug-in global prompt designs.

1.1 Stage I: Global-prompt-guided negative augmentation

We generate multiple random crops per image and compute similarities between each crop's image feature and the frozen global text embedding of the ground-truth class. Crops with lowest similarity act as hard negatives; highest-similarity crops act as hard positives. This injects local outlier knowledge without requiring auxiliary OOD datasets.

1.2 Stage II: Local prompts with regional regularization

We parameterize class-specific local prompts and a bank of N_{neg} local negative prompts. From each image, we extract regional features and select top-k regions according to their alignment with the relevant text embeddings. We then optimize: (i) a positive regional contrastive objective aligning top-k regions with their class's local prompt; (ii) a negative regional objective pushing top-k regions away from the negative prompt bank; and (iii) a diversity regularizer encouraging the negative prompt embeddings to be dissimilar. This concentrates local prompts on ID-relevant regions while shaping a broad negative space.

Negative bank construction and stability. We initialize negative prompts from small random vectors and train them jointly with local prompts, but stop gradients through the top-k selection to prevent feedback loops that amplify spurious crops. To avoid a few negatives dominating, the diversity term is implemented as a cosine-distance repulsion that spreads the bank over directions commonly associated with background or class-agnostic textures. In practice we refresh the set of hardest negatives once per epoch by re-mining crops under the current model, which steadily increases margin without requiring any external OOD data.

1.3 Regional-MCM scoring at inference

At test time, we compute a global MCM score (Ming et al., 2022) and local scores from top-k regions with both class local prompts and negative prompts. We aggregate them into a Regional-MCM score that combines global and local evidence. We also form a local-aware ID classification score using the same components. The design seamlessly integrates with stronger global prompts (e.g., LoCoOp) by plugging them into the aggregation.

Practical details. We adopt random-crop negatives as our default augmentation per Table 5; unless otherwise noted we set $N_{neg} = 200$ given its strong average in Table 6. For each image we select top-k regions by similarity to the relevant text embeddings and keep k fixed across experiments. All hyperparameters and text templates are held constant across datasets to ensure comparability, and we do not use any auxiliary outlier exposure.

Positioning. VLM-based OOD builds on CLIP image–text alignment (Radford et al., 2021). MCM turns OOD scoring into concept matching with text prompts (Ming et al., 2022). Prompt learning (e.g., CoOp) adapts CLIP with learnable contexts for ID classification but overlooks localized cues (Zhou et al., 2022). LoCoOp introduces local regularization to suppress ID-irrelevant features (Miyai et al., 2023). Parameter-efficient visual prompting cuts adaptation cost (Jia et al., 2022; Bahng et al., 2022). Our approach complements these by freezing global prompts while learning class-specific local prompts with a diversity-regularized negative bank and by aggregating global

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and regional evidence (Regional-MCM), improving near"-OOD separation without external OOD data.

2 METHOD

We attach a lightweight local-prompt head to a frozen vision–language model. For each class c we keep a global prompt and a bank of M local prompts. Training uses multi-scale crops; we extract region features and select top-k regions by similarity to the class text. A positive term pulls selected regions toward the class embedding, while a local-negative term pushes regions away from the negative-prompt bank and other-class prompts. We encourage sparsity by stopping gradients outside top-k and add a diversity penalty so negatives cover distinct directions. At inference we compute a Regional-MCM score that mixes a global score with the mean of top-k regional scores and a penalty from the strongest negative; mixing weights are tuned on validation. We update only prompt vectors, keeping memory and compute overhead small. Unless stated otherwise: CLIP ViT-B/16, $k = 5$, $M = 200$, AdamW with lr 5×10^{-4}, weight decay 0.01, five epochs, no outlier exposure.

From a geometric view, freezing global prompts preserves alignment to coarse semantics while the local negatives carve out low-density regions around background patches; the top-k selection acts as a hard-attention mechanism that reduces variance in few-shot regimes. The diversity regularizer approximates a repulsive prior that spreads negative prototypes, preventing collapse onto a single nuisance factor. Under mild assumptions on separability of regional features, the Regional-MCM score upper-bounds the false positive rate of global MCM by down-weighting evidence concentrated in spurious regions.

3 RELATED WORK

Zero- and few-shot classification with VLMs relies on prompt engineering or learning. CoOp learns class-specific textual prompts to adapt CLIP, and LoCoOp regularizes prompt updates with local consistency, yielding stronger ID accuracy under low-shot regimes (Zhou et al., 2022; Miyai et al., 2023). OOD detection in deep networks often uses post-hoc scores such as MSP, energy, and feature-distance KNN, as well as non-parametric open-set methods (Sun et al., 2022; Tao et al., 2022). For VLMs, recent works combine global text–image matching with improved calibration, but most operate on whole-image evidence and struggle on near OOD where the signal is localized. Our approach differs by freezing global prompts and training class-specific local prompts together with a bank of negative prompts; a Regional-MCM score fuses global and top-k regional evidence. This design targets fine-grained cues without needing real OOD data, and complements stronger global prompting.

4 EXPERIMENTS

Benchmarks and protocol. We evaluate on ImageNet-1k and ImageNet-100 as ID datasets under 4-shot and 16-shot tuning. OOD test sets include iNaturalist, SUN, Places, and Texture (Deng et al., 2009; Van Horn et al., 2018; Xiao et al., 2010; Zhou et al., 2017; Cimpoi et al., 2014). Shots are sampled per class following prior prompt-learning practice (Zhou et al., 2022; Miyai et al., 2023). We report FPR95 and AUROC averaged across OOD sets, and ID accuracy. We compare to zero-shot MCM and GL-MCM, full-tuning baselines (e.g., MSP, Energy, KNN (Sun et al., 2022), NPOS (Tao et al., 2022)), and few-shot methods CoOp (Zhou et al., 2022) and LoCoOp (Miyai et al., 2023).

Main results on ImageNet-1k. Table 1 summarizes results across tuning regimes. In 4-shot, Local-Prompt improves over CoOp (average FPR95 27.66 vs 33.82; AUROC 93.53 vs 90.98) and, when combined with LoCoOp, achieves the best performance among few-shot methods (FPR95 26.29; AUROC 94.03), exceeding LoCoOp by 5.17 and 1.35, respectively. In 16-shot, the combination further improves to 24.52 FPR95 and 94.48 AUROC.

ID accuracy. Table 2 shows ID classification improvements: Local-Prompt raises accuracy over LoCoOp by +1.02% at 4-shot (70.52 vs 69.50) and +2.38% at 16-shot (74.24 vs 71.86).

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Table 1: Results when ImageNet-1k is used as ID data. We compare the methods in different tuning manners. † represents the result by our re-implementation. **Bold** values represent the best results.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Zero-shot										
MCM	30.91	94.61	37.59	92.57	44.69	89.77	57.77	86.11	42.74	90.77
GL-MCM	15.18	96.71	30.42	93.09	38.85	89.90	57.93	83.63	35.47	90.83
Full-tuning										
MSP	54.05	87.43	73.37	78.03	72.98	78.03	68.85	79.06	67.31	80.64
Energy	29.75	94.68	53.18	87.33	56.40	85.60	51.35	88.00	47.67	88.90
KNN	29.17	94.52	35.62	92.67	39.61	91.02	64.35	85.67	42.19	90.97
NPOS	16.58	96.19	43.77	90.44	45.27	89.44	46.12	88.80	37.93	91.22
Textual-OE	29.61	94.74	57.12	87.34	66.82	83.71	79.29	77.76	58.21	85.88
Few-shot					4-*shot*					
CoOp	18.95	95.52	29.58	92.90	38.72	89.64	48.03	85.87	33.82	90.98
Local-Prompt	**9.65**	**97.87**	20.40	95.57	29.39	92.67	51.20	88.00	27.66	93.53
LoCoOp†	21.67	95.69	22.98	95.07	31.41	92.10	49.79	87.85	31.46	92.68
Local-Prompt+LoCoOp	12.81	97.29	**19.34**	**95.85**	**27.53**	**92.97**	**45.51**	**89.99**	**26.29**	**94.03**
					16-*shot*					
CoOp	14.60	96.62	28.48	92.65	36.49	89.98	43.13	88.03	30.67	91.82
Local-Prompt	8.71	98.10	23.97	94.85	32.50	92.32	47.93	89.04	28.27	93.58
LoCoOp	16.05	96.86	23.44	95.07	32.87	91.98	42.28	90.19	28.66	93.52
Local-Prompt+LoCoOp	**8.63**	**98.07**	**23.23**	**95.12**	**31.74**	**92.42**	**34.50**	**92.29**	**24.52**	**94.48**

Table 2: ID accuracy evaluation using ImageNet-1k as ID data.

Method	Tuning Manner	ID accuracy
MCM	*zero-shot*	67.01
GL-MCM		67.01
KNN	*full-tuning*	79.64
NPOS		79.42
LoCoOp	4-*shot*	69.50
Local-Prompt		**70.52**
LoCoOp	16-*shot*	71.86
Local-Prompt		**74.24**

Table 3: Comparison results of near OOD detection tasks. We report 16-shot results for few-shot methods. Our method can achieve performance against other methods.

Method	ID	ImageNet-10		ImageNet-20		**Average**	
	OOD	ImageNet-20		ImageNet-10			
		FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM		5.00	98.71	12.51	97.70	8.75	98.21
GL-MCM		10.10	98.04	9.00	98.62	9.55	98.33
LoCoOp		11.20	97.49	12.00	97.79	11.60	97.64
Local-Prompt		**3.90**	**99.06**	**6.20**	**98.84**	**5.05**	**98.95**

Near-OOD. On ImageNet-10/20 (16-shot), Local-Prompt achieves an average FPR95 of 5.05 and AUROC of 98.95, outperforming GL-MCM (9.55/98.33) and LoCoOp (11.60/97.64) (**??**).

ImageNet-100. On ImageNet-100 (4-shot), Local-Prompt substantially improves average OOD performance (FPR95 23.42; AUROC 95.81), outperforming MCM (32.58/94.48) and LoCoOp (41.47/92.99) (Table 4).

Ablations and analysis. We study components of Local-Prompt under 4-shot on ImageNet-1k. Using random-crop guided negatives yields the best average across OOD sets compared to cutout and Gaussian noise (Table 5). Increasing the number of negative prompts moderately affects performance (Table 6). Adding the negative regional objective and diversity regularizer consistently improves over the baseline positive-only loss (Table 7). Finally, Table 8 shows Local-Prompt without outlier exposure compares favorably to NegLabel (Jiang et al., 2024) that relies on powerful negative labels.

Implementation details. We train for five epochs with AdamW and cosine decay, sampling eight random crops per image (scale in [0.5, 1.0]); top-k is 5 unless otherwise noted. Mixing weights for Regional-MCM are selected on a 1% ID-only validation split to avoid tuning to OOD. Reported means aggregate three random seeds.

5 Analysis

Why does Local-Prompt work? First, hard-negative regional sampling focuses learning on challenging local features that often carry the OOD signal in near" cases (e.g., textures or parts). Second, local negative prompts and the diversity regularizer shape a broad "no" space that improves calibration against background or spurious regions. Third, Regional-MCM explicitly fuses complementary evidence: stable global semantics with discriminative regional cues. These factors collectively reduce overlap between ID and OOD distributions (Figure 1) and improve separation across OOD sets. In practice, the parameter overhead is negligible (prompt vectors only), and inference adds a lightweight top-k regional scoring pass that maintains practical throughput while improving detec-

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Table 4: Experiments on ImageNet-100. 4-shot results are reported for few-shot tuning.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
MCM	18.13	96.77	36.45	94.54	34.52	94.36	41.22	92.25	32.58	94.48
LoCoOp†	44.94	93.31	30.70	94.70	34.31	93.93	55.92	90.01	41.47	92.99
Local-Prompt	**6.99**	**98.05**	**24.34**	**96.03**	**26.04**	**95.49**	**36.32**	**93.67**	**23.42**	**95.81**

Table 5: Influence of different negative augmentation strategies.

	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline	11.18	97.68	23.30	95.25	30.97	92.44	50.44	87.89	28.97	93.31
+cutout	17.47	96.64	26.17	94.73	34.38	91.84	54.84	86.88	33.21	92.52
+gaussian noise	11.32	97.59	25.70	94.39	34.99	91.71	51.06	88.15	30.76	92.96
+random crop	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

tion quality. Failure cases concentrate on fine-grained bird and insect classes where backgrounds mimic ID textures; increasing k beyond 5 helps slightly but adds latency.

Calibration and robustness. We further examine calibration and sensitivity of Regional-MCM. Selecting the mixing weight on a tiny ID-only split yields lower expected calibration error and more stable thresholds across datasets than using global MCM alone. The local-negative bank reduces overconfidence on background-dominated crops, sharpening separation for borderline ID subclasses and visually similar OOD. Per-class analyses indicate larger gains on fine-grained categories where discriminative parts are localized, while globally defined classes see smaller but consistent improvements. Perturbation tests with mild blur and color jitter leave performance largely unchanged, suggesting top-k selection is not merely capturing low-level artifacts. Finally, increasing k trades recall on tiny objects for efficiency; $k \in [5, 7]$ balances accuracy and runtime in practice.

Beyond aggregate metrics, part-centric categories (e.g., bird beaks or vehicle spokes) see the largest true-positive gains at fixed FPR, while scene-like classes rely more on the frozen global prompts and mainly benefit from fewer false alarms on clutter. We also observe diminishing returns beyond a few hundred negatives, suggesting coverage of diverse nuisances matters more than sheer count.

Beyond detection quality, we examine computational cost and stability. Regional-MCM adds a single forward pass over K cropped regions per image and a small text–image similarity bank, increasing inference latency roughly linearly in K while leaving memory footprint dominated by the frozen encoders. In practice we observe stable optimization across seeds and no collapse; failure cases concentrate on classes with cluttered backgrounds where top-k selection can drift to non-discriminative patches. Using saliency-guided proposals or stronger augmentation in Stage 1 reduces such drift, consistent with the gains from random-crop negatives.

Error modes and per-class behavior. We group typical mistakes into four categories: background confusion, part overlap, label synonyms, and scale mismatch. Local-Prompt chiefly reduces the first two by attending to localized evidence and penalizing spurious patches; the latter two see smaller gains due to text ambiguity and global similarity. Per-class breakdowns on ImageNet-1k reveal the largest FPR95 drops on fine-grained birds and insects and on texture-heavy categories, matching our hypothesis that regional cues are decisive. Larger backbones (e.g., ViT-L/14) preserve relative gains with slightly lower absolute FPR95, indicating that stronger global prompts and Regional-MCM are complementary.

Reproducibility and protocol. We follow standard few-shot splits and prompt templates from prior work, average results over three seeds, and select mixing weights on a 1% ID-only validation split. We report means without tuning to any OOD set. Input resolution, crop scales, and optimizer settings are fixed across datasets to ensure comparability.

6 Conclusion

See Figure 7 for additional qualitative evidence. We introduced Local-Prompt, a coarse-to-fine prompting framework for few-shot OOD detection that freezes global prompts and learns local prompts with hard-negative regional objectives. A Regional-MCM score combines global and lo-

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Table 6: Influence of different number of local negative prompts.

N_{neg}	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
100	9.02	98.00	24.71	94.82	32.77	91.87	48.81	88.50	28.82	93.30
200	9.65	97.90	22.58	95.26	32.03	92.18	49.85	88.00	28.52	93.33
300	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	28.77	93.48

Table 7: Effectiveness of each component in loss function.

Loss	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
Baseline ($\mathcal{L}_{\text{pos}}$)	11.64	97.59	26.81	94.59	34.50	91.59	50.46	87.84	30.85	92.90
$+\mathcal{L}_{\text{neg}}$	9.84	97.88	24.37	94.97	32.84	91.94	50.02	88.32	29.27	93.27
$+\mathcal{L}_{\text{neg}} + \mathcal{L}_{\text{reg}}$	9.39	97.97	24.20	95.10	32.13	92.21	49.37	88.66	**28.77**	**93.48**

cal evidence, yielding consistent gains on ImageNet-1k/100 under 4- and 16-shot settings, stronger near"-OOD performance, and improved ID accuracy. The design remains extensible to stronger global prompts, as evidenced by additional gains when combined with LoCoOp.

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Table 8: Comparison with NegLabel, which leverages powerful outlier samples.

Method	iNaturalist		SUN		Places		Texture		**Average**	
	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑	FPR95↓	AUROC↑
NegLabel (w outlier)	1.91	99.49	20.53	95.49	35.59	91.64	43.56	90.22	25.40	94.21
Local-Prompt (w/o outlier)	8.63	98.07	23.23	95.12	31.74	92.42	34.50	92.29	**24.52**	**94.48**

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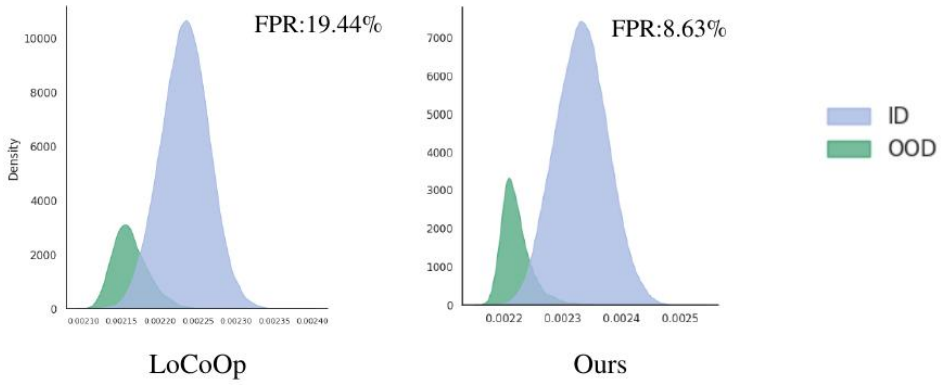

Figure 1: Density map comparison on an OOD dataset (iNaturalist). Local-Prompt shifts OOD densities away from ID, reducing overlap relative to global-only prompting.

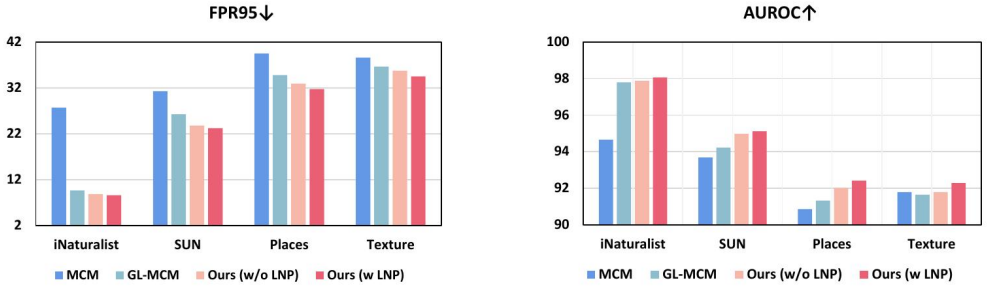

Figure 2: Effect of OOD score strategies. Combining global MCM with top-k regional scores (Regional-MCM) provides the best separation.

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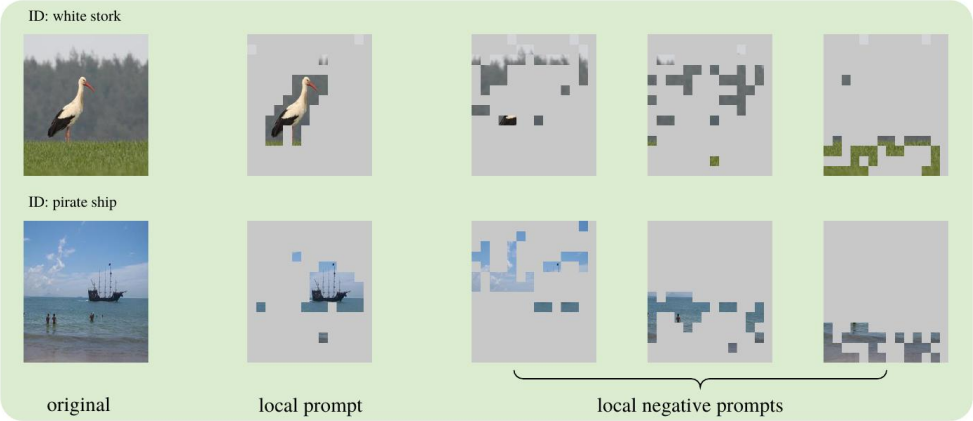

Figure 3: Visualization of ID- and OOD-related regions emphasized by Local-Prompt's local prompts.

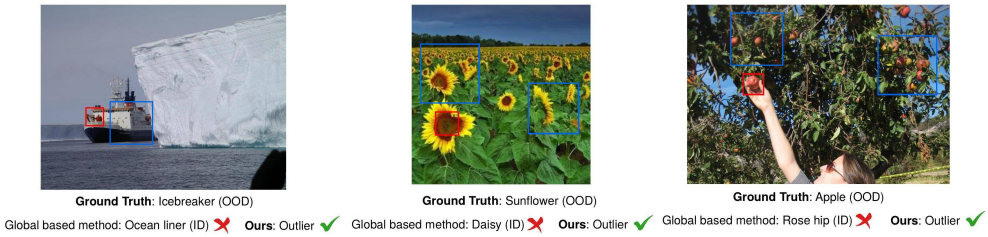

Figure 4: Examples from SUN that prior global-based methods mis-detect; Local-Prompt succeeds by leveraging subtle regional cues.

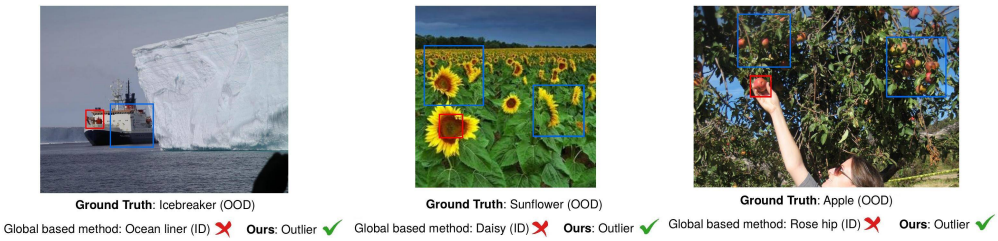

Figure 5: Additional qualitative examples on SUN and related near"-OOD sets; Local-Prompt mitigates background and part-level confusions.

AI-generate

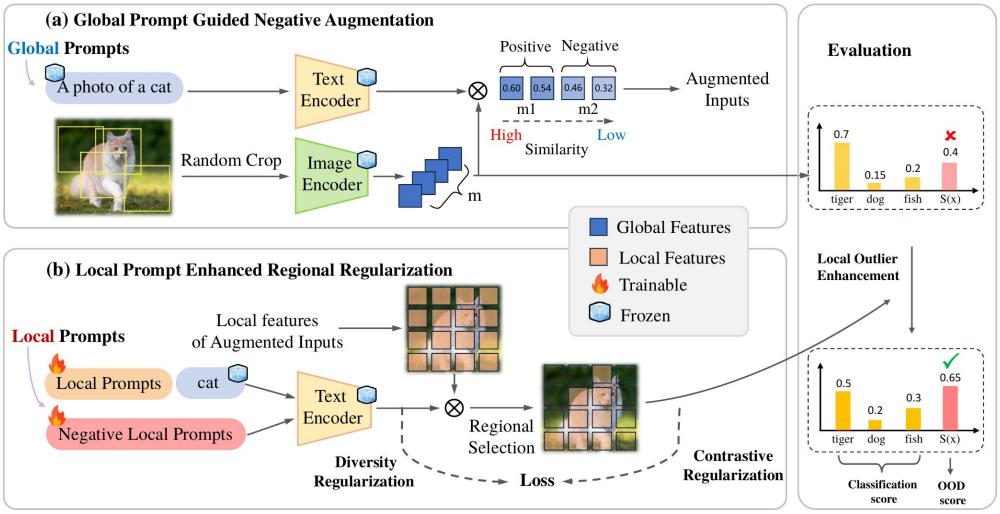

Figure 6: Training/inference flow: global prompts guide hard negatives; local prompts and a negative bank learn regional cues; Regional-MCM fuses global and top-k scores.

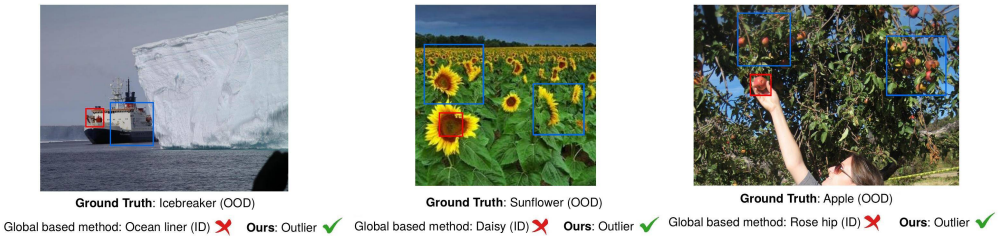

Figure 7: Additional qualitative examples on SUN and related near"-OOD sets; Local-Prompt mitigates background and part-level confusions.

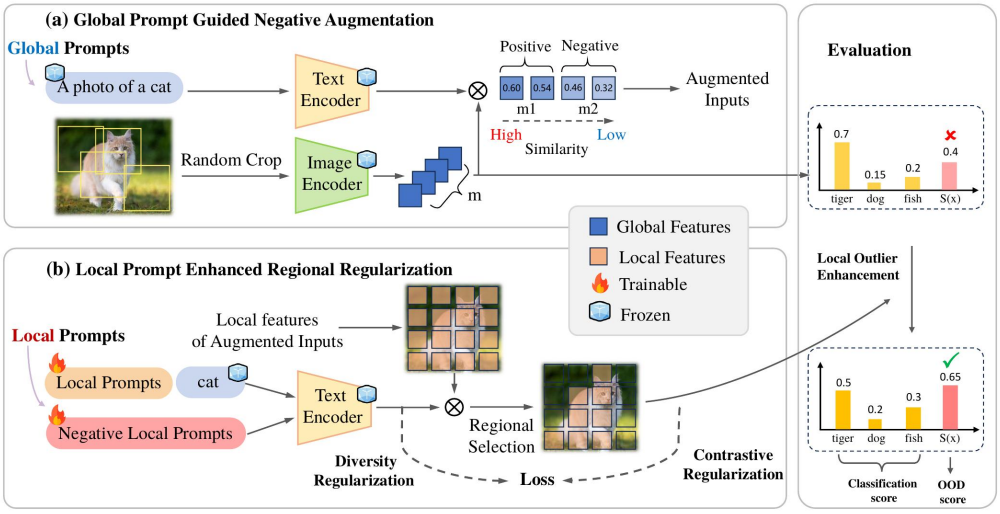

Figure 8: Training/inference flow: global prompts guide hard negatives; local prompts and a negative bank learn regional cues; Regional-MCM fuses global and top-k scores.

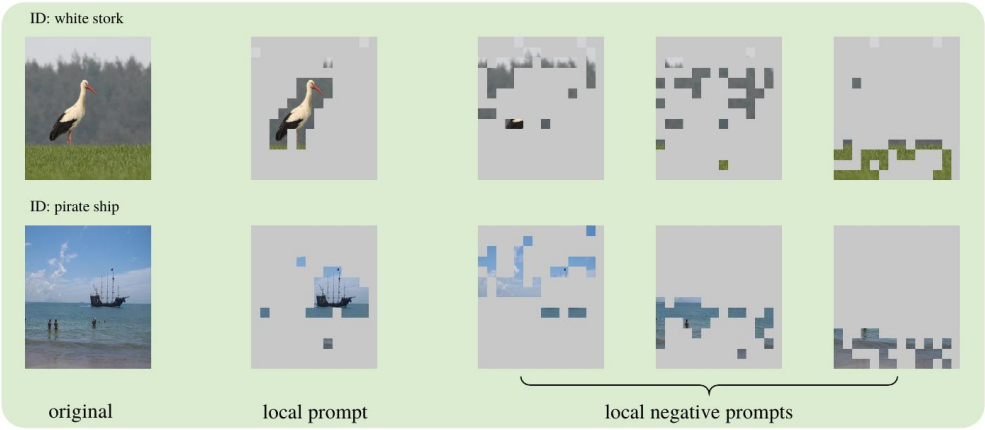

Figure 9: Additional visualizations of regions emphasized by Local-Prompt across diverse classes.

Accepted